Mock exam 1

Planetary Systems final exam practice • closed book • 120 minutes

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This mock exam has the same shape and level as the final exam: 6 questions of 15 points each, 90 points in total, in a 120-minute closed-book sitting where a calculator and a pen are all that is allowed. Your grade is 1 + points/10; the pass mark of 6.0 corresponds to 50 points. The twelve relations on the exam formula list are assumed known; every other formula a question needs is printed in the question itself. Marks per part are shown in brackets, and the steps carry marks, not only the number: show your algebra. Some parts state an intermediate result ("show that …"); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps and round only at the end.

Constants and data

$G = 6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ 1 AU = 1.496×10^{11} m 1 yr = 3.156×10^{7} s = 365.25 d 1 d = 86 400 s $M_\odot = 1.989 \times 10^{30}$ kg $R_\odot = 6.957 \times 10^{8}$ m R_J = 71 492 km $k_B = 1.381 \times 10^{-23}\ \mathrm{J\,K^{-1}}$ $m_u = 1.661 \times 10^{-27}$ kg $\sigma = 5.670 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$ $S_0 = 1361\ \mathrm{W\,m^{-2}}$ (solar constant at 1 AU)

Body	a, A	Further data
Earth	1.000 AU, A = 0.30	mean surface heat flux 0.092 $\mathrm{W\,m^{-2}}$
Venus	0.723 AU, A = 0.77	surface T = 737 K, surface P = 92 bar, g = 8.87 $\mathrm{m\,s^{-2}}$, μ = 43.4, c_p = 1130 $\mathrm{J\,kg^{-1}\,K^{-1}}$
Mars	1.524 AU, A = 0.25	mean surface T = 215 K, atmospheric D/H 6 times Earth's, present surface water ≈ 25 m global equivalent layer
Io		$M = 8.93 \times 10^{22}$ kg, R = 1821 km, tidal heat output 1×10^{14} W

A total power L spread over a sphere of radius r gives a flux $L/(4\pi r^2)$; for sunlight this means $S = S_0/a^2$ with a in AU. Present-day radiogenic heating rate of chondritic rock: $5 \times 10^{-12}\ \mathrm{W\,kg^{-1}}$. Half-life of ^{40}K: 1.25 Gyr. Age of the solar system: 4.567 Gyr.

Problem 1 Route to Venus

Treat the orbits of Earth and Venus as circular and coplanar. A spacecraft transfers between them on an ellipse whose aphelion touches Earth's orbit and whose perihelion touches Venus's orbit. At each end the spacecraft fires its engine briefly (a *burn*); a burn along the direction of motion (*prograde*) speeds it up, a burn against it (*retrograde*) slows it down. Work entirely in the Sun's gravitational field and ignore the gravity of the planets themselves. The vis-viva equation gives the speed at distance r on an orbit with semi-major axis a:

$$v^2 = GM_\odot\left(\frac{2}{r} - \frac{1}{a}\right).$$

(a) **[2 pts]** Compute the orbital period of Venus, in days.

(b) **[4 pts]** Compute the semi-major axis of the transfer ellipse, and show that the flight to Venus takes 146.0 days.

(c) **[4 pts]** The spacecraft starts at the aphelion of the transfer ellipse, on Earth's orbit. Compute its speed there and the speed of Earth on its own orbit, and state the magnitude and direction (prograde or retrograde) of the departure burn.

(d) **[3 pts]** Launch opportunities repeat with the *synodic period* S, the time between successive identical alignments of Earth and Venus. Starting from the angular speeds $2\pi/P$ of the two planets, show that

$$\frac{1}{S} = \frac{1}{P_\mathrm{V}} - \frac{1}{P_\mathrm{E}},$$

and compute S in days.

(e) **[2 pts]** At launch, must Venus lead or trail Earth along its orbit for the spacecraft to meet it at arrival? Compute the required angle.

Problem 2 Beneath the clouds of Venus

The surface conditions of Venus and the properties of its CO_2-dominated atmosphere are listed in the data table.

(a) **[3 pts]** Assume the atmosphere near the surface is isothermal at the surface temperature. Show that its pressure scale height is $H = 15.9$ km.

(b) **[4 pts]** In this isothermal approximation the pressure falls with altitude as $P(z) = P_0\, e^{-z/H}$. At what altitude does the pressure fall to 1 bar? The 1 bar level on Venus actually sits near 50 km. Explain in 1 to 2 sentences why the isothermal estimate overshoots.

(c) **[4 pts]** Below the clouds the temperature profile of Venus is close to dry adiabatic, with lapse rate

$$\Gamma = \frac{g}{c_p}.$$

Compute Γ and the temperature at 50 km altitude.

(d) **[4 pts]** Judge this claim with a supporting calculation, in 3 to 4 sentences: *"Venus is hotter than Earth at the surface because its clouds absorb more sunlight."*

Problem 3 Io's heat engine

Io, the innermost Galilean moon of Jupiter, is the most volcanically active body in the solar system. Its mass, radius, and measured heat output are listed in the data table.

(a) **[2 pts]** Compute Io's mean density. Is Io made mostly of rock or of ice?

(b) **[3 pts]** Compute Io's mean surface heat flux and compare it with Earth's.

(c) **[3 pts]** Explain in 2 to 3 sentences where Io's heat comes from, and why the mechanism would shut down if Io were the only large moon of Jupiter.

(d) **[4 pts]** Judge this claim with a supporting calculation, in 2 to 3 sentences: *"Io's volcanism is powered by radioactive decay, like the interior heat of Earth."*

(e) **[3 pts]** The Moon has nearly the same size and density as Io, yet shows no volcanic activity today. Explain the difference in 2 to 3 sentences.

Problem 4 Reading time from rock

Planetary surfaces are dated in two ways: relatively, by crater counting, and absolutely, by radiometric dating of samples. The decay law and the half-life relation are on the formula list; the half-life of ^{40}K is in the data table.

(a) **[3 pts]** Explain in 3 to 4 sentences how crater densities order planetary surfaces in time, and what anchors the *absolute* ages of lunar surfaces.

(b) **[5 pts]** A lunar mare basalt contains 7 times as many ^{40}Ar atoms as ^{40}K atoms. Assume the rock has retained all argon since it solidified, and that every decayed ^{40}K atom became ^{40}Ar (in reality about 11% do; ignore this here). Show first that one eighth of the original ^{40}K remains, then compute the age of the rock.

(c) **[3 pts]** By what factor was a planet's ^{40}K inventory larger at the formation of the solar system than it is today? State in 1 sentence what this implies for the heat budgets of young planets.

(d) **[4 pts]** A volcanic plain on Mars has the same crater density as the lunar mare from part (b). A fellow student concludes that both surfaces are 3.75 Gyr old. Judge this conclusion in 2 to 3 sentences.

Problem 5 The climate history of Mars

Orbital and surface data for Mars are listed in the data table. Geological evidence (valley networks, lake beds) shows that liquid water flowed on early Mars.

(a) **[3 pts]** Show first that the stellar constant at Mars is $S = 586\ \mathrm{W\,m^{-2}}$, then compute the equilibrium temperature of Mars and compare its present greenhouse warming with Earth's 33 K.

(b) **[4 pts]** Deuterium (D) is twice as heavy as ordinary hydrogen (H) and escapes to space far less easily. Assume early Mars started with the same D/H ratio as Earth today. Use the measured D/H enrichment to estimate the minimum depth of the water layer early Mars possessed, and explain in 1 to 2 sentences why your estimate is a lower limit.

(c) **[4 pts]** On Earth, the carbonate-silicate cycle stabilises the climate over millions of years. Explain in 3 to 4 sentences how this feedback works, and why it failed on Mars.

(d) **[4 pts]** A water-rich planet cannot radiate more than roughly 280 to 310 $\mathrm{W\,m^{-2}}$ from a moist atmosphere (the runaway greenhouse limit). Judge this claim with a supporting calculation, in 3 to 4 sentences, keeping Mars's albedo unchanged: *"If Mars orbited at Venus's distance, it would undergo a runaway greenhouse."*

Problem 6 A planet from a light curve

A transit survey observes a Sun-like star ($M_* = 1.00\ M_\odot$, $R_* = 1.00\ R_\odot$). The star's brightness dips by 1.00%, and the dip repeats every 3.50 days. The transit depth δ measures the sky-projected area ratio:

$$\delta = \left(\frac{R_\mathrm{p}}{R_*}\right)^2 .$$

(a) **[3 pts]** Compute the radius of the transiting object, in Jupiter radii.

(b) **[4 pts]** Show that the orbital semi-major axis is $a = 0.0451$ AU.

(c) **[4 pts]** Compute the stellar constant at the planet's orbit and, assuming a Bond albedo of $A = 0.10$, its equilibrium temperature. What kind of planet is this?

(d) **[4 pts]** Judge this claim in 3 to 4 sentences: *"From this light curve alone we know that the object is a gas giant planet."*